

Avinash Tiwari, Ph.D. Student

✉ itsmeavinash@gmail.com
✉ avinash.tiwari@iucaa.in
🌐 <https://avinash-tiwari-at.github.io/>
🆔 0000-0001-7197-8899



Current Affiliations

- 2021 – 📌 Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2022 – 📌 LIGO-Virgo-Kagra (LVK) Collaboration and LIGO-India Scientific Collaboration (LISC).

Education

- 2021 – 📌 **Ph.D.**, Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2019 – 2021 📌 **M.Sc. Physics**, Indian Institute of Technology Guwahati, Guwahati.
- 2016 – 2019 📌 **B.Sc. Physics, Mathematics, and Computer Science**, University of Allahabad, Prayagraj.

Research Interests

- 📌 Bayesian inference and GW data analysis, especially, but not restricted to, Fisher Matrix Analysis, Parameter Estimation, and Hierarchical (Hyperparameter) Inference.
- 📌 GW waveform modeling, especially line-of-sight acceleration (LOSA) waveforms
- 📌 Probing the compact binary coalescence (CBC) environments using GWs.
- 📌 GW population studies.
- 📌 Cosmology.

Skills

- Languages 📌 English, Hindi.
- Coding 📌 Python, C, $\text{\LaTeX}$.
- Web Dev 📌 GITHUB PAGES.
- Packages 📌 BILBY, BILBY_PIPE, BILBY_TGR, GWPOPULATION, EMCEE, DYNESTY.

Accomplishments

- 📌 **Graduate Aptitude Test in Engineering** (GATE 2021): All India Rank 209 in Physics.
- 📌 **Junior Research Fellowship** (JRF)-CSIR: All India Rank 89 in Physical Sciences, Joint CSIR - UGC NET JUNE 2020 (held in November 2020).
- 📌 **IASC** (November 9 – December 4, 2020): Member of the team “Delhi University” in the “DST-Rajasthan Asteroid Search Campaign” organised by DST-Rajasthan in collaboration with the International Astronomical Search Collaboration (IASC).
- 📌 **IASC** (August 12 – September 8, 2020): member of the team “IIT Guwahati” in the “DST-Rajasthan Asteroid Search Campaign” organised by DST-Rajasthan in collaboration with the International Astronomical Search Collaboration (IASC).
- 📌 **Institute Merit-cum-Means** (McM) Scholarship: awarded in both years of M.Sc. (2019-21) by the India Institute of Technology Guwahati on merit-cum-means basis for being in the top 25%.

Accomplishments (continued)

- Joint Admission Test for M.Sc. (JAM) 2019: All India Rank 236 in Physics.

Conferences

- GR24 - Amaldi16** (July 14- 18, 2025): Talk on “Profiling stellar environments of gravitational wave sources”.
- ICGC-2023** (December 6 – December 9): Talk on “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- Amaldi15** (July 17- 21, 2023): Talk on “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- IAGRG32** (December 19 - 21, 2022): Poster presentation and a Flash Talk on “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

Workshops and Seminars

- Last Friday Talks at IUCAA** (August 29, 2025): “Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries”.
- Last Friday Talks at IUCAA** (September 27, 2024): “Profiling stellar environments of gravitational wave sources”.
- AEI Colloquium** (August 22, 2024), AEI Hanover: “Profiling stellar environments of gravitational wave sources”.
- New ideas on the ORIGIN of BLACK HOLE MERGERS** (August 12 – August 16, 2024), NBI Copenhagen: “Talk on Profiling stellar environments of gravitational wave sources”.
- Last Friday Talks at IUCAA** (July 28, 2023): “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- LUNAR GRAVITATIONAL-WAVE DETECTION** (April 17 - 20, 2023), Ramanujan Lecture Hall, ICTS Bengaluru.
- Last Friday Talks at IUCAA** (March 31, 2023): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.
- Online Seminar at ICTS, Bengaluru** (December 15, 2022): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

Courses Tutored

- PHY-IC481 - General Relativity at IUCAA.
- PHY-IE485 - Elective course - Cosmology at IUCAA.

Research Publications

Short author works

- A. Tiwari**, S. A. Bhat, S. J. Kapadia, T. R. Choudhury, S. Adhikari, and M. K. Singh, “Inference of 21-cm cosmic hydrogen signal parameters using information from Gravitational Waves,” 2025, In preparation.
- A. Tiwari**, S. A. Bhat, M. A. Shaikh, and S. J. Kapadia, “Testing the nature of GW200105 by probing the frequency evolution of eccentricity,” 2025, Undergoing PnP review.
- A. Tiwari**, P. Chanda, S. J. Kapadia, S. Adhikari, A. Vijaykumar, and B. Dasgupta, “Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries,” Aug. 2025, Under review in PRL. arXiv: 2508.03803 [hep-ph].

- 4 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, and S. Chatterjee, “Identifying and characterizing extragalactic circum-CBC exoplanets with future gravitational-wave detectors,” 2025, In preparation.
- 5 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, S. Ghosh, and A. B. Nielsen, “A pipeline to search for signatures of line-of-sight acceleration in gravitational wave signals produced by compact binary coalescences,” Jun. 2025, Under review in PRD. arXiv: 2506.22272 [astro-ph.HE].
- 6 S. A. Bhat, **A. Tiwari**, M. A. Shaikh, and S. J. Kapadia, “EECT: an Eccentricity Evolution Consistency Test to distinguish eccentric gravitational-wave signals from eccentricity mimickers,” Aug. 2025, Under review in PRD. arXiv: 2508.14850 [gr-qc].
- 7 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, S. Chatterjee, and G. Fragione, “Profiling stellar environments of gravitational wave sources,” Jul. 2024, Under review in PRD. arXiv: 2407.15117 [astro-ph.HE].
- 8 **A. Tiwari**, A. Vijaykumar, S. J. Kapadia, G. Fragione, and S. Chatterjee, “Accelerated binary black holes in globular clusters: forecasts and detectability in the era of space-based gravitational-wave detectors,” *Mon. Not. Roy. Astron. Soc.*, vol. 527, no. 3, pp. 8586–8597, 2023.  DOI: 10.1093/mnras/stad3749. arXiv: 2307.00930 [astro-ph.HE].
- 9 A. Vijaykumar, **A. Tiwari**, S. J. Kapadia, K. G. Arun, and P. Ajith, “Waltzing Binaries: Probing the Line-of-sight Acceleration of Merging Compact Objects with Gravitational Waves,” *Astrophys. J.*, vol. 954, no. 1, p. 105, 2023.  DOI: 10.3847/1538-4357/acd77d. arXiv: 2302.09651 [astro-ph.HE].

Collaboration Papers

- 1 A. G. Abac et al., “All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA Observing run,” Jul. 2025. arXiv: 2507.12282 [gr-qc].
- 2 A. G. Abac et al., “All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run,” Jul. 2025. arXiv: 2507.12374 [astro-ph.HE].
- 3 A. G. Abac et al., “Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run,” Sep. 2025. arXiv: 2509.07352 [gr-qc].
- 4 A. G. Abac et al., “GW231123: a Binary Black Hole Merger with Total Mass 190-265 $M_{\odot}$,” Jul. 2025. arXiv: 2507.08219 [astro-ph.HE].
- 5 A. G. Abac et al., “GW250114: Testing Hawking’s Area Law and the Kerr Nature of Black Holes,” *Phys. Rev. Lett.*, vol. 135, no. 11, p. 111403, 2025.  DOI: 10.1103/kw5g-d732. arXiv: 2509.08054 [gr-qc].
- 6 A. G. Abac et al., “GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog,” *arXiv e-prints*, arXiv:2508.18080, Aug. 2025, Art. no. arXiv:2508.18080.  DOI: 10.48550/arXiv.2508.18080. arXiv: 2508.18080 [gr-qc].
- 7 A. G. Abac et al., “GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients,” Aug. 2025. arXiv: 2508.18081 [gr-qc].
- 8 A. G. Abac et al., “GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run,” Aug. 2025. arXiv: 2508.18082 [gr-qc].
- 9 A. G. Abac et al., “Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run,” Aug. 2025. arXiv: 2508.18079 [gr-qc].
- 10 A. G. Abac et al., “Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run,” *Astrophys. J.*, vol. 983, no. 2, p. 99, 2025.  DOI: 10.3847/1538-4357/adb3a0. arXiv: 2501.01495 [astro-ph.HE].
- 11 A. G. Abac et al., “Search for Gravitational Waves Emitted from SN 2023ixf,” *Astrophys. J.*, vol. 985, no. 2, p. 183, 2025.  DOI: 10.3847/1538-4357/adc681. arXiv: 2410.16565 [astro-ph.HE].

- 12 A. G. Abac et al., “Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA’s fourth Observing Run,” Aug. 2025. arXiv: 2508.20721 [gr-qc].
- 13 “Black Hole Spectroscopy and Tests of General Relativity with GW250114,” Sep. 2025. arXiv: 2509.08099 [gr-qc].
- 14 T. L. S. Collaboration, the Virgo Collaboration, and the KAGRA Collaboration, “Gwtc-4.o: Constraints on the cosmic expansion rate and modified gravitational-wave propagation,” 2025. arXiv: 2509.04348 [astro-ph.CO].  URL: <https://arxiv.org/abs/2509.04348>.
- 15 “GW230814: investigation of a loud gravitational-wave signal observed with a single detector,” Sep. 2025. arXiv: 2509.07348 [gr-qc].
- 16 “GWTC-4.o: Population Properties of Merging Compact Binaries,” Aug. 2025. arXiv: 2508.18083 [astro-ph.HE].
- 17 G. Raman et al., “Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO–Virgo–KAGRA Observing Run,” *Astrophys. J.*, vol. 980, no. 2, p. 207, 2025.  DOI: 10.3847/1538-4357/ad9749. arXiv: 2407.12867 [astro-ph.HE].
- 18 A. G. Abac et al., “A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154,” *Astrophys. J.*, vol. 977, no. 2, p. 255, 2024.  DOI: 10.3847/1538-4357/ad8de0. arXiv: 2410.09151 [astro-ph.HE].
- 19 A. G. Abac et al., “Observation of Gravitational Waves from the Coalescence of a 2.5–4.5 $M_{\odot}$ Compact Object and a Neutron Star,” *Astrophys. J. Lett.*, vol. 970, no. 2, p. L34, 2024.  DOI: 10.3847/2041-8213/ad5beb. arXiv: 2404.04248 [astro-ph.HE].
- 20 A. G. Abac et al., “Ultralight vector dark matter search using data from the KAGRA O3GK run,” *Phys. Rev. D*, vol. 110, no. 4, p. 042001, 2024.  DOI: 10.1103/PhysRevD.110.042001. arXiv: 2403.03004 [astro-ph.CO].